

Day 11-12 – Assignment For SQL Section

ABC Insurance Ltd. wants to digitize its policy management process. Currently everything is maintained in Excel files. You have been hired as a Database Developer. Your responsibility is to design the database and populate it with sample data. Create a Notepad/Word document for performing actions below:

1. Design a database named **InsuranceDB**. The system should maintain information about:
 - Customers
 - Insurance Policies
 - Agents
 - Policy Purchases
2. Create tables considering schema given and with appropriate constraints like Primary Key, Unique Key, Foreign Key, Check constraint and Default value if required.

Customer

Column	Datatype
CustomerId	INT Identity
CustomerName	VARCHAR(100)
Gender	CHAR(1)
Phone	VARCHAR(15)
Email	VARCHAR(100)
City	VARCHAR(50)

Agent

Column	Datatype
AgentId	INT Identity
AgentName	VARCHAR(100)
Phone	VARCHAR(15)
Branch	VARCHAR(50)

Policy

Column	Datatype
PolicyId	INT Identity
PolicyName	VARCHAR(100)
PolicyType	VARCHAR(50)
PremiumAmount	DECIMAL(10,2)
CoverageAmount	DECIMAL(12,2)

PolicyPurchase

Column	Datatype
PurchaseId	INT Identity
CustomerId	INT
PolicyId	INT
AgentId	INT
PurchaseDate	DATE
PolicyTerm	INT

3. Draw the ER Diagram and Identify
 - Entities
 - Attributes
 - Primary Keys
 - Foreign Keys
 - Relationships
4. Insert sample data which should look a like actual data. NO GARBAGE TEXT. Minimum 15 Customers, 5 Agents and 6 Policies, try with 20+ records in PolicyPurchase.
5. Examples
 - Health Insurance
 - Car Insurance
 - Bike Insurance
 - Home Insurance
 - Travel Insurance
 - Life Insurance

Write SQL Scripts for the following:

1. Display all details of every customer.
2. Display only the following columns from the Customer table:
 - Customer Name
 - Phone
 - City
3. Display all insurance policies whose premium amount is greater than ₹20,000.
4. Display all customers who belong to **Hyderabad**.
5. Display all policies in ascending order of premium amount.
6. Display the **Top 3** most expensive insurance policies.
7. Display all customers whose names start with the letter **A**.
8. Display all agents working in the **Delhi** branch.
9. Display all policies whose premium amount is between ₹15,000 and ₹30,000.
10. Display customers whose city is either:
 - Hyderabad
 - Chennai
 - Pune
11. Display all policies except **Life Insurance**.
12. Display customers whose phone number ends with **55**.
13. Display the following information:
 - Customer Name
 - Policy Name
 - Purchase Date
14. Display the following information:
 - Customer Name
 - Policy Name
 - Purchase Date
15. Display all customers who purchased **Life Insurance**.
16. Display all policies sold by **each agent**. Expected columns:
 - Agent Name
 - Policy Name
17. Display customer details along with the premium amount of the policy they purchased.
 - Expected columns:
 - Customer Name
 - Policy Name
 - Premium Amount
18. Display the total number of policies sold by each agent. Expected columns:
 - Agent Name
 - Number of Policies Sold
19. Display the customer who has purchased the maximum number of policies. Expected columns:
 - Customer Name
 - Total Policies Purchased
20. Display the average premium amount for all insurance policies.
21. Display the policy with the highest premium.
22. Display the policy with the lowest premium.

23. Count the total number of customers in each city.
24. Count how many policies exist for each policy type.
25. Display all customers who have not purchased any policy.
26. Display all agents who have never sold a policy.
27. Display the total premium amount sold by each agent.
28. Display all policy purchases made in the current year.
29. Display customers who have purchased more than one policy.
30. Display the oldest policy purchase based on the purchase date.

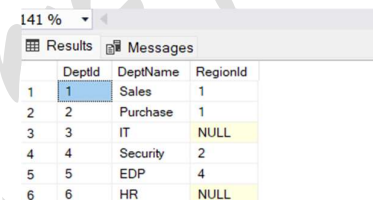
Each trainee should submit:

- SQL script to create the database
- SQL script to create all tables
- SQL script to insert sample data
- SQL script containing UPDATE statements
- SQL script containing DELETE statements
- SQL script containing all SELECT queries
- A hand-drawn or digital ER diagram

NOTE: Prepare one MS Word document with SQL commands fired (SQL Script) by you and also partial screen-shot of output obtained to prove successful execution of your command as:

Example Query: Display all records of Dept table

Answer SQL Script: **select * from dept;**



	DeptId	DeptName	RegionId
1	1	Sales	1
2	2	Purchase	1
3	3	IT	NULL
4	4	Security	2
5	5	EDP	4
6	6	HR	NULL

☞*☞ Debug your doubts, deploy your success ☞*☞